

HARSH KESHARI

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EDUCATION

KIET Group of Institutions	Ghaziabad, India
Bachelor of Technology, Computer Science & Engineering (AI)	Aug 2021 – Jun 2025

SKILLS

Languages & Frameworks: TypeScript, JavaScript, Python, Node.js, Express, React, Next.js, Prisma
AI/ML: OpenAI (GPT-4o, embeddings), Deepgram (Nova-3, speaker diarization), Pinecone vector search, Vertex AI, Google Cloud Speech-to-Text
Cloud, Infra & Integrations: AWS (EC2, Lambda), Vercel, Docker, GCP, Firebase, Bull + Redis queues, node-cron, Google Calendar/Meet, Recall.ai, Razorpay, Brevo
Databases & Architecture: PostgreSQL, MongoDB, Redis, Pinecone; Microservices, Multi-tenancy, Vector Search

EXPERIENCE

Founding Engineer	Nov 2024 – Present
Experiment Labs	<i>Remote</i>

- **Architected recommendation engine** on OpenAI embeddings + Pinecone vector search — primary revenue driver, adopted by Leverage Edu and PhysicsWallah; driving personalized activity recommendations for student career roadmaps.
- **Launched 10 production microservices** across ExperimentLabs’ edtech platform — core (Auth, CRM, Payments, Notifications), AI services (Recommendation Engine, Smart Meeting Assistant, AI Interviewer), and product modules (Internship Portal, Evaluator, Sales Operations Dashboard).
- **Designed central SSO/identity service** (auth-core: 92 Prisma models, 85 routes) powering all microservices — refactored permission model from 93 → 38 across 6 categories; multi-tenant org isolation via tenant-keyed queries.
- **Engineered Smart Meeting Assistant** — counselor-student session recording with AI summaries auto-pushed to student career roadmaps; Bull + Redis queues + Railway worker processing batch transcription across orgs.
- **Developed AI Interviewer end-to-end** — Google Cloud Speech-to-Text for student answers, Vertex AI for adaptive question generation and scoring, async video transcoding via Cloud Tasks.
- **Implemented Payments and Notifications services** — Razorpay subscription billing with webhook reconciliation and Brevo transactional email with node-cron scheduled reminders.
- **Migrated long-running services (transcription, video processing, recommendation regeneration) from AWS Lambda to EC2**, eliminating serverless timeouts; added cron monitors for payment reconciliation and meeting-link retries.
- **Delivered 13+ React/Next.js frontends end-to-end** — counselor CRM, AI Interviewer portal, ops dashboard, embeddable plugin, and marketing site, deployed alongside each backend service.

OPEN SOURCE

Oppia Foundation (Quality Team Lead, Nov 2023 – Apr 2024) — led LaCE, Oppia’s frontend quality team; merged 10 PRs across admin acceptance tests and UI fixes.
Tailcall — resolved search modal focus bug; earned \$150 bounty from the maintainer team.
Also contributed to **CircuitVerse** (digital circuit simulation) and **Zowe** (Linux Foundation’s open mainframe platform).

PROJECTS

Crelyzor — crelyzor.app | **PERN stack · TypeScript · Prisma · Bull/Redis · Next.js · Deepgram Nova-3 · Google Gemini 2.0 Flash · Recall.ai · Resend · Docker · GitHub Actions**

- **All-in-one productivity SaaS for solo professionals** — replaces HiHello (cards) + Cal.com (scheduling) + Otter.ai (meeting AI) + Todoist (tasks). Live in production with billing, multi-environment Docker deploys, and a public booking widget embeddable on any site.
- **Engineered end-to-end meeting intelligence pipeline** — Deepgram Nova-3 Multilingual transcription with speaker diarization, Google Gemini 2.0 Flash for summaries + AI-extracted tasks; per-meeting Ask AI with SSE streaming + persisted conversation history; Recall.ai bots auto-join Meet/Zoom on bookings and feed the same pipeline.
- **Implemented timezone-aware scheduling engine** — slot calculation across availability schedules, GCal busy-time, existing bookings, and buffers; serializable transactions prevent double-booking races; bidirectional GCal push webhooks for real-time sync.
- **Structured 4-repo production system** — API backend, authenticated dashboard, SSR public site (cards, booking, shared meetings), and separate admin portal; Docker multi-stage builds, 3-environment Compose, nginx+Certbot SSL, GitHub Actions selective deploys.
- **Launched billing infrastructure** — Free/Pro (\$19)/Business tiers, per-resource usage metering (transcription minutes, Recall hours, AI credits, storage), 402 enforcement with in-context upgrade flow, monthly reset cron.
- **Hardened production security** — encryption at rest via Google Cloud KMS (per-user DEKs, AES-256-GCM for transcripts/tasks/contacts), GCS bucket CMEK, crypto-shredding on account delete (GDPR); fixed XSS, SSRF on OG route; admin JWT in httpOnly cookies, CSP, rate-limited admin auth.